

Real Analysis HW 7

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A function f is said to be additive if $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. Prove that if f is continuous at some point x_0 , then it is continuous at every point of $\mathbb{R}$

First I will prove some intermediate properties of f

- $f(x) = f(x + 0) = f(x) + f(0)$
Thus $f(0) = 0$
- $f(0) = f(x - x) = f(x) + f(-x)$
Thus $-f(x) = f(-x)$
- Also $f(c) = f(c - x_0) + f(x_0)$

We have that f is continuous at x_0 , since x_0 must be a cluster point, since all points $\in \mathbb{R}$ are cluster points of $\mathbb{R}$, then

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

$$\begin{aligned}\lim_{x \rightarrow x_0} f(x) &= \lim_{x \rightarrow x_0} [f(x - x_0) + f(x_0)] \\ &= \lim_{x \rightarrow x_0} [f(x - x_0)] + f(x_0) \\ &= f(x_0)\end{aligned}$$

$$\Rightarrow \lim_{x \rightarrow x_0} f(x) = 0 = f(0)$$

If $\alpha = x - x_0$

$$\Rightarrow \lim_{\alpha \rightarrow 0} f(\alpha) = f(0)$$

Therefore f is continuous at 0

Let $c \in \mathbb{R}$, and $\epsilon > 0$

Then

$$|f(x) - f(c)| = |f(x - c)|$$

Letting $\beta = x - c$

Since $f(\beta)$ is continuous at 0 and $f(0) = 0$, we have

$$\forall \epsilon > 0 \exists \delta \text{ such that } 0 < |\beta| < \delta \Rightarrow |f(\beta)| < \epsilon$$

then

$$\forall \epsilon > 0 \exists \delta \text{ such that } 0 < |x - c| < \delta \Rightarrow |f(x - c)| < \epsilon$$

Thus $f(c)$ is continuous $\forall c \in \mathbb{R}$

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Let $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfy the relation $g(x+y) = g(x)g(y)$ for all $x, y \in \mathbb{R}$. Show that if g is continuous at $x = 0$, then g is continuous at every point of $\mathbb{R}$. Also if we have $g(a) = 0$ for some $a \in \mathbb{R}$, then $g(x) = 0$ for all $x \in \mathbb{R}$

$g(x)$ is continuous at $x = 0$

Thus $\lim_{x \rightarrow 0} g(x) = g(0)$

In all of what follows, we assume $g(x) \neq 0 \forall x$

$$\begin{aligned} g(x) &= g(x+0) = g(x)g(0) \\ &\Rightarrow g(0) = 1 \end{aligned}$$

Also

$$\begin{aligned} 1 &= g(0) = g(x + (-x)) = g(x)g(-x) \\ &\Rightarrow g(x) = \frac{1}{g(-x)} \end{aligned}$$

Then, let $c \in \mathbb{R}$

$$\begin{aligned} \lim_{x \rightarrow c} g(x) = g(c) &\iff \lim_{x \rightarrow c} \frac{g(x)}{g(c)} = 1 \\ &\iff \lim_{x \rightarrow c} \frac{g(x)}{\left(\frac{1}{g(-c)}\right)} = 1 \\ &\iff \lim_{x \rightarrow c} g(x - c) = 1 \\ &\iff \lim_{\beta \rightarrow 0} g(\beta) = 1 \end{aligned}$$

Which is true since, we have that g is continuous at 0

Thus $g(x)$ is continuous $\forall x \in \mathbb{R}$, under the assumption $g(x) \neq 0 \forall x$

Let $c \in \mathbb{R}$

If $\exists a \in \mathbb{R}$ s.t. $g(a) = 0$, then

$$\begin{aligned} g(c) &= g(c - a + a) = g(c - a)g(a) = g(c - a)(0) = 0 \\ &\Rightarrow g(c) = 0 \forall c \in \mathbb{R} \end{aligned}$$

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Let $I := [a, b]$ and let $f : I \rightarrow \mathbb{R}$ be a continuous function on I such that for each $x \in I$ there exists $y \in I$ such that $|f(y)| \leq \frac{1}{2}|f(x)|$. Prove that there exists a point $c \in I$ such that $f(c) = 0$

Since we are dealing with a function f which is continuous on a closed interval we know that there will be both a maximum value, say $f(t)$, and minimum value, say $f(s)$, in the interval (Extreme Value Theorem)

First I will show that $f(t) \geq 0$

Assume that $f(t) < 0$,

Then there exists $y \in I$, s.t. $|f(y)| \leq \frac{1}{2}|f(t)|$

If $f(y) \leq 0$, then from the above property, its absolute value is less than that of $f(t)$ and thus, $f(y) > f(t)$, a contradiction

If $f(y) > 0$, then $f(y) > f(t)$, a contradiction

Thus $f(t) \geq 0$

Now we look at $f(s)$

Assume $f(s) > 0$

Then there exists $y \in I$, s.t. $|f(y)| \leq \frac{1}{2}|f(s)| = \frac{1}{2}f(s)$, a contradiction since this implies $f(y) < f(s)$

Thus $f(s) \leq 0$

Now we have it that $f(t) \geq 0 \geq f(s)$, then by the Intermediate Value Theorem, there exists $c \in I$ s.t. $f(c) = 0$

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Show that the polynomial $p(x) := x^4 + 7x^3 - 9$ has at least two real roots. Use a calculator to locate these roots to within two decimal places

We see that $p(-10) > 0$ and $p(-5) < 0$, then Intermediate Value Theorem $\Rightarrow$ on the closed interval $[-10, -5]$, p has at least one root

We see that $p(0) < 0$ and $p(5) > 0$, then Intermediate Value Theorem $\Rightarrow$ on the closed interval $[0, 5]$, p has at least one root

Then on $p : \mathbb{R} \rightarrow \mathbb{R}$, p has at least two roots

For the first root, we can shrink the interval down to $[-7.026, -7.025]$ (i.e. the signs of the endpoints of the interval are different), thus to two decimal places the root is given by -7.03

For the second root, we can shrink the interval down to $[1.037, 1.039]$, thus to two decimal places the root is given by 1.04